Introduction to Python Programming
(NES103/NES-203)

Course Outcomes:

1. Develop the ability to solve various computational problems efficiently using Python, paving the way for application development and data analysis.

Unit	Topic
I	**Introduction:** Basic data types - Strings, Lists, Tuples - Lists and Tuples - Strings and Unicode strings Buffers - Dictionaries - Numbers - Type conversions - Files. **Syntax rules:** Indentation - Line structure - Block structure - Special objects.
II	**Variables and namespaces:** Variables - Multiple assignments - Assignments, references and copies of objects - Namespaces - Accessing namespaces. Control flow: Conditionals - Loops - while - for - more about loops.
III	**Functions:** Operators - Order of evaluation - Object comparisons - dot operator - String formatting - Defining functions - Passing arguments to parameters - Reference arguments - Passing arguments by keywords - Default values of parameters - Variable number of parameters.
IV	**Exceptions:** General Mechanism - Python built-in exceptions - Raising exceptions - Defining exceptions. Modules and packages: Modules - Loading - Packages – Loading
V	**Classes:** Creating instances - Getting information on a class.

Reference Books:

1. Mark Lutz David Ascher (2003) Learning Python. O'Reilly & Associates
2. Alan Gauld (2000) Lean To program Using Python Addison -Wesley